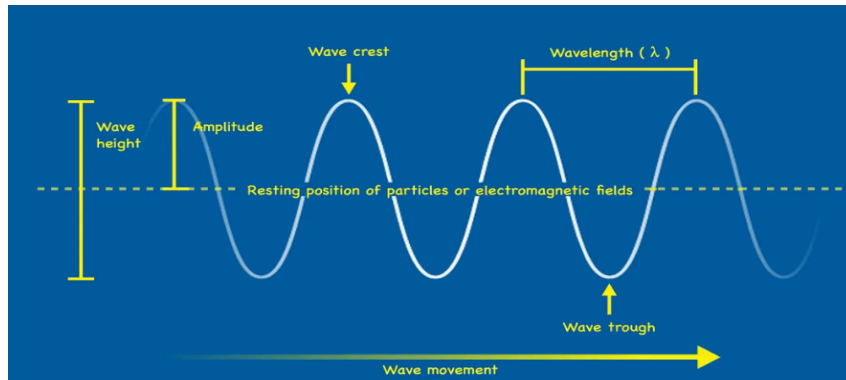


Sc4S2U2 Tier 2 Knowledge & Skills Assessment

VOCABULARY



WORD	KHMER or CHINESE	MEANING
Wave	រលក 波	A disturbance that moves energy from one place to another.
Rest Position	ទីតាំងគ្មានរលក 平衡位置	The position of the medium when there is no wave or disturbance.
Crest	កំពូលរលក 波峰	The highest point of a wave.
Trough	បាតរលក 波谷	The lowest point of a wave.
Amplitude	អំពូទុយត 振幅	The height of a wave from the rest position to the crest.
Frequency	ប្រេកង់ 频率	How many waves pass a point in a certain amount of time.
Pattern	លំនាំ 模式 / 规律	A repeated design or sequence in waves.
Reflect	ចំណាំងផ្លាត 反射	When a wave bounces off a surface.
Absorb	ស្រូប 吸收	When a wave's energy is taken in by a material.
Pass through	ឆ្លងកាត់ 透过 / 穿过	When a wave moves all the way through a material.

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Part A – Understanding Waves (K1, K2, K3, S1, S2, S3)

1. Which statement best describes a *wave*?

តើ Statement មួយណាដែលពិពណ៌នាអំពីរលកបានល្អបំផុត?

- A. A flow of water moving downhill
- B. A pattern of motion that carries energy
- C. The up-and-down motion of an object
- D. A solid object moving through space

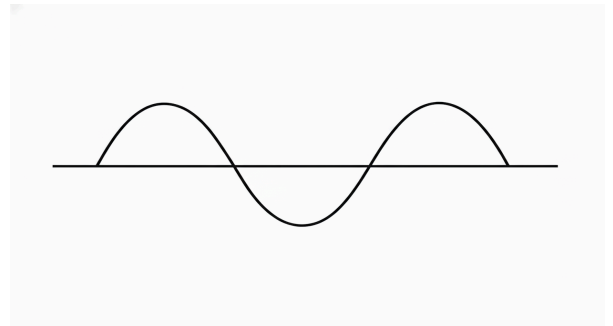
2. Look at the wave on the rope.

មើលរលកនៅលើខ្សែក្នុងរូបភាព

- a. Label the crest (A) and trough (B) on the drawing.

- b. Draw and label:

- an arrow to show the wavelength (from one crest to the next crest), and
- an arrow to show the amplitude (from the middle line up to the crest).



3. Fill in the blank with the words provided below.

បំពេញចន្លោះដោយពាក្យដែលបានផ្តល់ខាងក្រោម។

wavelength	amplitude	higher	lower
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- a. The distance between two crests is the _____.
- b. A taller wave has a _____ amplitude.
- c. Waves with _____ amplitude carry more energy than waves with _____ amplitude.

4. Tick (✓) whether each statement about seismic waves is true or false.

គូស(✓) ថាតើ Statement នីមួយៗអំពីរលករញ្ជួយដីគឺពិតឬមិនពិត។

Statement	True	False
Seismic waves travel through Earth's surface.	☐	☐

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They can cause buildings to move or shake.	☐	☐
They do not follow any pattern.	☐	☐
They carry sound through the air.	☐	☐

Part B – Light and Sound (K4, S2, S4)

5. **Match** the object to what happens when light hits it.

ផ្គូផ្គងវត្ថុទៅនឹងអ្វីដែលកើតឡើងនៅពេលពន្លឺប៉ះវា។

Object			What Happens to Light
Mirror	☐	☐	Light reflects
Window	☐	☐	Light passes through
Wall	☐	☐	Light is absorbed
Shiny metal	☐	☐	Light bounces back clearly

6. A student stands in a dark room with a shiny ball.



The student turns on a flashlight and can now see the ball.

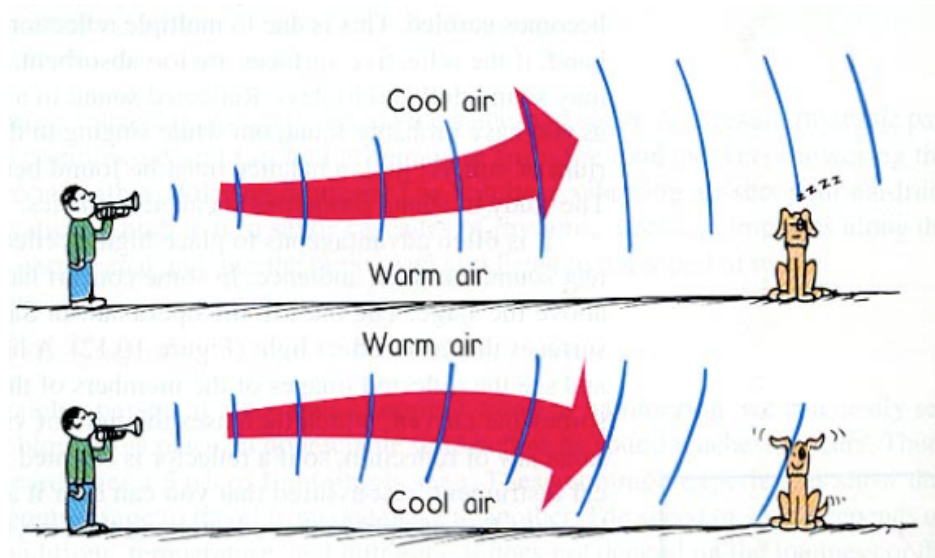
Explain why the ball can now be seen.

7. Complete the table by ticking (✓) what happens when sound waves hit a surface.

បំពេញតារាងដោយគូស (✓)

តើមានអ្វីកើតឡើងនៅពេលរលកសំឡេងប៉ះលើផ្ទៃ។

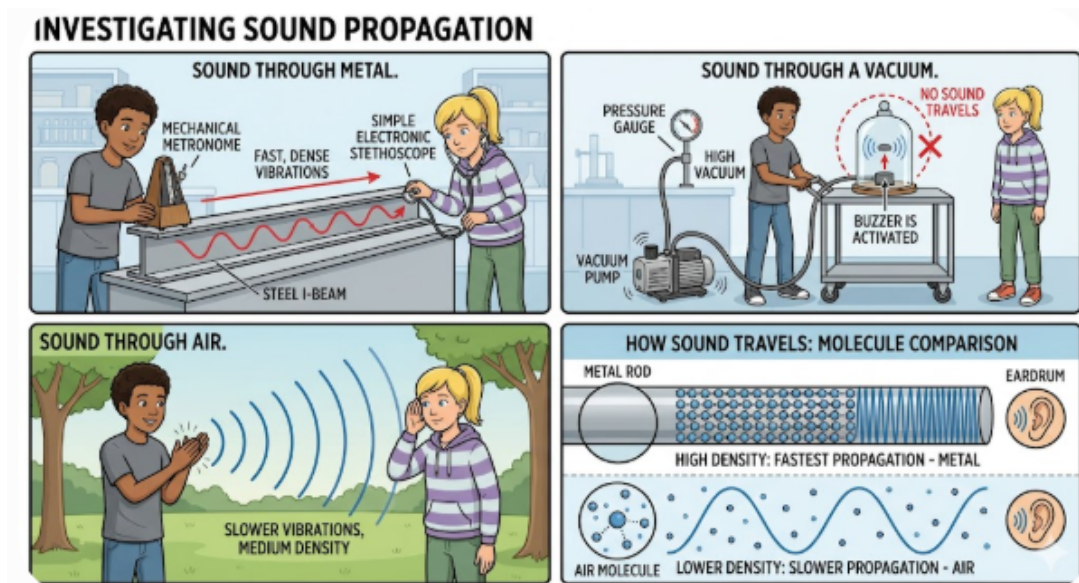
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Surface	Does sound reflect, absorb, or pass through?		
Wall	☐ Reflect	☐ Absorb	☐ Pass through
Curtain	☐ Reflect	☐ Absorb	☐ Pass through
Metal	☐ Reflect	☐ Absorb	☐ Pass through

8. Two students test sound in different materials.

សិស្សពីរនាក់សាកល្បងសំឡេងនៅក្នុងសម្ភារៈផ្សេងគ្នា។



Material	Sound Heard (Yes/No)
Air	Yes
Metal	Yes

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Vacuum	No
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What pattern do you notice about how sound travels?

តើអ្នកសម្គាល់ឃើញគំរូអ្វីអំពីរបៀបដែលសំឡេងធ្វើដំណើរ?

Part C – Waves for Communication (K5, S4, S5)

9. Which is an example of using patterns to send information?

តើមួយណាជាឧទាហរណ៍នៃការប្រើលំនាំដើម្បីផ្ញើព័ត៌មាន?

- A. Talking to a friend
- B. Writing on paper
- C. Flashing a light in Morse code
- D. Playing a song on an instrument



10. Match each signal to the way it sends information.

ផ្គូផ្គងសញ្ញានីមួយៗទៅនឹងវិធីដែលវាបញ្ជូនព័ត៌មាន។

Signal Type

Pattern Used

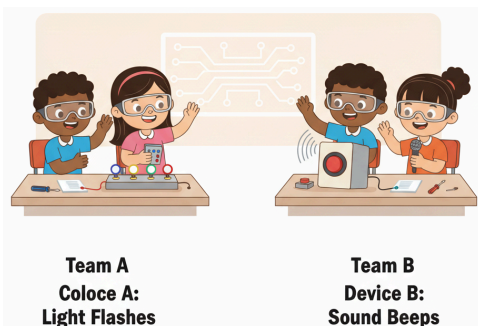
- | | | |
|------------------|--------------------------|--|
| Sound drum beats | ☐ | ☐ Loud/soft rhythm patterns |
| Light flashes | ☐ | ☐ On/off patterns |
| Radio waves | ☐ | ☐ Electrical frequency patterns |
| Digital text | ☐ | ☐ Code of numbers and letters |

11. Two student teams make communication devices.

Both need to send a message across a noisy playground.

ក្រុមសិស្សពីរនាក់បង្កើតឧបករណ៍ទំនាក់ទំនង។

អ្នកទាំងពីរត្រូវផ្ញើសារឆ្លងកាត់សួនកុមារដែលមានសំឡេងរំខាន។



Team A
Coloce A:
Light Flashes

Team B
Device B:
Sound Beeps

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Which design will likely work better, and why?

តើការចនាមួយណាទំនងជានឹងដំណើរការល្អជាង ហើយហេតុអ្វី?

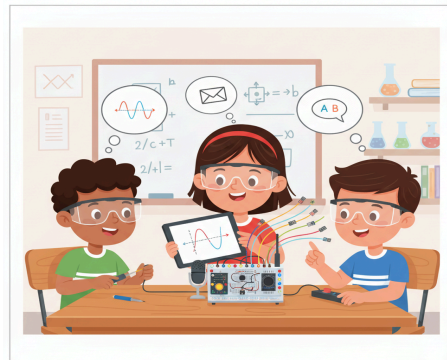
12. Students design a device to send information using waves.

សិស្សរចនាឧបករណ៍ដើម្បីបញ្ជូនព័ត៌មានដោយប្រើរលក។

Circle the best idea that meets the criteria:

- Must work in daylight
- Must not be too loud

- A. A whistle
- B. A flashlight signal
- C. A drum
- D. A bell



Rubric

_____/12

Part A – Waves and Motion (K1–K3, S1–S3)		Standards: 4-PS4-1
Items 1–4		
4	Accurately identifies wave parts, patterns, and properties (amplitude, wavelength); shows clear reasoning in models or tables.	
3	Identifies most features correctly; shows general understanding with minor errors.	
2	Attempts but includes incomplete or confused reasoning.	
1	Mentions parts or terms with incorrect explanation.	
0	No attempt or no understanding demonstrated.	

Part B – Light and Sound (K4, S2, S4)	Standards: 4-PS4-2
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Items 5–8	
4	Explains how light reflection and sound travel allow sight and hearing; accurately completes tables or reasoning tasks.
3	Identifies most examples correctly; simple but logical reasoning.
2	Attempts but includes partial or inaccurate explanations.
1	Gives vague or unrelated responses.
0	No attempt or no understanding demonstrated.

Part C – Communication and Design (K5, S4, S5) Items 9–12		Standards: 4-PS4-3, 3-5-ETS1-2
4	Accurately identifies, compares, and justifies design solutions using wave patterns and criteria/constraints.	
3	Identifies reasonable solutions with partial justification.	
2	Attempts comparison but reasoning unclear or incomplete.	
1	Mentions an idea but shows little link to design criteria.	
0	No attempt or no understanding demonstrated.	